

AI for Coders & Software Engineers

ENEC 2026 | Participant Handbook & Instructor Reference

Four-day applied enterprise AI programme

Databricks & Enterprise Data
Machine Learning & Forecasting
Generative AI & Retrieval-Augmented Generation
Agentic AI & Governed Workflows
Integrated A-001 Capstone

TRAINING BOUNDARY

All assets, documents, observations and data in this handbook are synthetic training material. The exercises teach enterprise AI design, evidence handling, governance and human oversight. They are not operational engineering instructions.

3.5 days guided learning + 0.5 day team capstone

Print-friendly working handbook: use it for class notes, labs, team planning and final project review.

How to Use This Handbook

This handbook is designed to serve two audiences at the same time: participants can use it as a working handout, while the instructor can use the marked reference boxes as a delivery guide. The programme follows one connected synthetic enterprise scenario rather than a different example every day.

For participants	For the instructor
Follow the day-by-day objectives, complete the lab checklists, write notes directly on the pages, and use the capstone templates with your team.	Use the INSTRUCTOR REFERENCE boxes as talking points, transitions and reminders. Keep the same A-001 story visible across all four days.
Use the quick-reference tables during Databricks exercises and the troubleshooting page if setup fails.	Protect the core outcomes if time runs short: setup, one forecast, one RAG flow, one SQL+RAG agent, one integrated capstone path.

Programme Outcomes

- Explore governed enterprise data in Databricks using SQL and Python.
- Build and interpret a simple A-001 vibration forecasting workflow.
- Explain practical LLM concepts and build a grounded RAG application.
- Distinguish relevant information from authoritative information.
- Build a governed agent that combines structured data and document knowledge.
- Explain tools, skills, knowledge, memory, progressive disclosure and multi-agent patterns.
- Integrate data, ML, RAG and agents into an evidence-to-human-review workflow.
- Demonstrate the final solution in a team capstone.

The Five Questions We Will Repeat

1. What do we know from the data?
2. What can we predict?
3. What does the organisation know?
4. What should the AI application do with that information?
5. Where must a human remain responsible?

INSTRUCTOR REFERENCE

Use a consistent teaching rhythm: WHY -> WHAT -> HOW -> SO WHAT. Every major concept should end by reconnecting to A-001 and the enterprise decision context.

Programme Roadmap

Phase	Core question	Primary hands-on output
Day 1 - Data & ML	What do we know, and what can we predict?	Working Databricks environment + A-001 vibration forecast
Day 2 - GenAI & RAG	What does the organisation know, and which source is authoritative?	Grounded A-001 knowledge assistant with sources
Day 3 - Agentic AI	Which approved tool should the application use next?	SQL + RAG agent with skills, state and governance
Day 4 first half - Integration	How do we combine evidence without losing provenance or control?	Integrated evidence-to-review workflow
Day 4 second half - Capstone	Can the team explain, demonstrate and govern the full solution?	Team demo + evidence-backed review briefing

Capability Journey

A-001 -> Databricks & Enterprise Data -> Machine Learning / Forecasting -> Generative AI -> Enterprise RAG -> AI Tools & Agents -> Skills + Memory + Progressive Disclosure -> Governed / Multi-Agent Workflows -> Integrated Enterprise AI Application -> Human Review & Oversight

Suggested Learning Rhythm

Block	What happens
Concept	Short explanation of the enterprise problem and the new capability.
Demo	Instructor shows the smallest working version first.
Hands-on	Participants run or adapt the notebook and inspect the output.
Interpretation	Class discusses what the output means - and what it does not prove.
Transition	A limitation in the current capability motivates the next topic.

KEY NARRATIVE We start with a table of sensor readings and progressively add prediction, organisational knowledge, tools, agents, workflow controls and human review. The technology changes; the A-001 story stays the same.
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Participant Setup Checklist

BEFORE CLASS

You need access to GitHub, Databricks and a modern browser. The instructor may pre-stage the environment, but every participant should understand the setup flow.

Repository

GitHub repository:

```
https://github.com/Decoding-Data-Science/enec2026
```

V2.2 Database

The training database is stored in the companion data package as:

```
nuclear_enterprise_360_v2_2_clean.db
```

Expected Databricks destination:

```
/Volumes/workspace/nuclear_enterprise_360/training_files/nuclear_enterprise_360_v2_2_clean.db
```

Minimum Successful Setup

Check	Ready when...
[]	Databricks workspace opens
[]	Repository notebooks are accessible
[]	V2.2 database is uploaded to the training_files Volume
[]	00_Setup_Nuclear_Enterprise_360.py runs successfully
[]	asset_360 returns A-001
[]	a001_sensor_hourly returns 8,760 rows
[]	Day 2 documents are discoverable
[]	Day 3 skill files are discoverable

Validation Queries

```
SELECT *
FROM workspace.nuclear_enterprise_360.asset_360
WHERE asset_id = 'A-001';

SELECT COUNT(*)
FROM workspace.nuclear_enterprise_360.a001_sensor_hourly;
-- Expected: 8,760
```

IF THE LIVE SETUP FAILS

Do not let setup consume the whole class. Use the pre-staged working environment, explain the import flow, and return to troubleshooting later. The learning objective is enterprise AI, not platform administration.

The A-001 Case: Our Single Enterprise Story

Throughout the programme we work with one synthetic asset: A-001, a cooling-water pump. We deliberately keep the scenario constant so every new AI capability can be understood as an extension of the same enterprise application.

Structured Evidence

- asset master and system context
- one-minute and hourly sensor readings
- asset health scores
- work orders
- inspections and maintenance history
- projects and project-to-asset relationships
- risks, actions and document links
- agent runs, tool calls and human approvals

Unstructured Evidence

#	Document	Status / role
01	Enterprise Asset Reliability & Escalation Policy	APPROVED
02	A-001 Operating & Usage Guide	APPROVED
03	Inspection Procedure V1	SUPERSEDED
04	Inspection & PM Procedure V2	APPROVED / current
05	Inspection Procedure V3D	DRAFT
06	Troubleshooting & Diagnostic Guide	APPROVED
07	Condition Monitoring Report	APPROVED
08	Inspection Work Order	OPEN
09	Field Condition Report	APPROVED

CORE GOVERNANCE TRAP

The newest or most semantically similar document is not automatically authoritative. V3D is newer but DRAFT; V2 remains the current APPROVED procedure. Enterprise retrieval must consider relevance AND authority.

Enterprise Data Map - What We Will Use

Object	Approx. rows	Teaching use
assets	128	Asset master
sensor_readings_a001_1min	525,600	Raw one-minute A-001 sensor history
a001_sensor_hourly	8,760	Primary Day 1 ML dataset
asset_health_scores	11,520	Condition deterioration
work_orders	1,500	Operational response / work context
inspections	600	Inspection evidence
maintenance_history	900	Maintenance history
projects	29	Enterprise project context
risks	423	Risk context
documents	42	Document register
a001_source_documents	9	A-001 RAG corpus metadata
agent_runs	50	Agent audit examples
agent_tool_calls	90	Tool-call traces
human_approvals	35	Human review examples

Views Used Repeatedly

View	Use
asset_360	Current asset summary for SQL/agent exercises
a001_sensor_hourly	Forecasting dataset
a001_event_timeline	Chronological A-001 story
asset_project_map	Asset-to-project relationships
asset_project_risk_map	Project and direct risk context
a001_document_evidence	A-001 document evidence relationships

REMEMBER

A database gives structured facts. Documents give organisational knowledge and context. Neither one replaces the other.

DAY 1

Databricks Foundations, Enterprise Data & Machine Learning

Set up the environment, explore A-001 data, and move from historical readings to a simple forecast.

Day 1 Learning Objectives

- Navigate a Databricks workspace without getting lost in platform detail.
- Understand catalog -> schema -> table / Volume.
- Load the portable SQLite training database into managed Delta tables.
- Explore A-001 using SQL and Python.
- Prepare hourly sensor data for a simple machine-learning workflow.
- Train, evaluate and interpret a vibration forecasting model.
- Explain why prediction alone is not enough for an enterprise maintenance decision.

Core Notebook Sequence

1. 00_Setup_Nuclear_Enterprise_360.py
2. 01_SQL_and_Delta_Foundations.py
3. A001_Forecasting_Databricks_Notebook.py

INSTRUCTOR REFERENCE

Open with the full A-001 capability journey before touching Databricks. The first half is about data foundations, not platform administration.

Day 1 - Databricks & SQL Quick Reference

The Setup Mental Model

```
Local V2.2 SQLite file
-> Unity Catalog Volume
-> Setup notebook
-> Managed Delta tables/views
-> SQL / Python / ML
```

Useful SQL

```
-- Find A-001
SELECT *
FROM assets
WHERE asset_id = 'A-001';

-- Current summary
SELECT *
FROM asset_360
WHERE asset_id = 'A-001';

-- Forecasting source
SELECT *
FROM a001_sensor_hourly
ORDER BY hour_timestamp
LIMIT 48;

-- Timeline
SELECT *
FROM a001_event_timeline
ORDER BY event_timestamp;
```

Questions to Ask While Exploring

- Which columns are observations, and which are derived summaries?
- What is the grain of the table - one asset, one hour, one event, one document?
- What joins are needed to move from an asset to work orders, projects, risks or documents?
- What does a NULL mean here - missing, not applicable, or not yet recorded?
- Which evidence is current state, and which is historical context?

INSTRUCTOR REFERENCE

Do not explain every line of setup code. Explain the purpose of each section: create location, upload file, convert tables, create useful views, validate, move on.

Day 1 - Machine Learning & Forecasting

Recommended Teaching Dataset

Item	Value
Dataset	a001_sensor_hourly
Rows	8,760 hourly observations
Primary target	avg_vibration_mm_s
Recommended horizon	7 days = 168 hours
Evaluation	MAE (Mean Absolute Error)

Simple ML Flow

```

Historical hourly data
-> data quality checks
-> visualise trends
-> define features and target
-> train / test
-> fit a simple regression model
-> calculate MAE
-> forecast 168 hours
-> interpret with enterprise context

```

MAE in Plain Language

MAE = average of $|\text{actual value} - \text{predicted value}|$. It tells us the typical absolute prediction error in the same units as the target.

Interpretation Questions

- Is the forecast directionally useful?
- How large is the typical error?
- Does the model capture the recent trend?
- What information is the model NOT using?
- Would you authorise maintenance from this forecast alone? Why not?

DAY 1 TRANSITION

Ask: Is prediction enough to make a maintenance decision? The answer should expose the missing organisational context - procedures, reports, work orders and authority. That gap becomes Day 2.

Day 1 - Hands-on Checklist & Notes

Done	Task
☐	Open Databricks and attach compute
☐	Run setup and validate V2.2 database
☐	Locate A-001 in asset_360
☐	Inspect a001_sensor_hourly
☐	Plot recent vibration trend
☐	Run the simple forecast
☐	Record MAE
☐	Write one sentence on what the model can tell us
☐	Write one sentence on what the model cannot tell us

My Day 1 Notes

DAY 2

Generative AI & Retrieval-Augmented Generation

Understand practical LLM behaviour, then ground answers in trusted A-001 enterprise knowledge.

Day 2 Learning Objectives

- Distinguish traditional ML from Generative AI.
- Explain prompts, tokens, context and model parameters at a practical level.
- Recognise hallucination and unsupported-answer risk.
- Explain chunking, embeddings and semantic retrieval.
- Build an A-001 RAG pipeline.
- Use metadata, approval status and versioning to improve trust.
- Produce grounded answers with sources.

OPENING QUESTION

Ask a general LLM: Which inspection procedure is currently approved for A-001? The model cannot reliably know. That limitation motivates RAG.

Day 2 - LLM Foundations in Practical Terms

Concept	Practical meaning
Prompt	The instruction and context sent to the model.
Token	A unit of text used by the model; token limits affect context size and cost.
Context window	The amount of information available to the model for one interaction.
Temperature	Controls output variability; lower values are usually preferred for bounded enterprise tasks.
Structured output	A predictable schema such as JSON, useful for downstream systems.
Hallucination	A fluent answer that is unsupported, invented or incorrectly grounded.
Foundation model	A general-purpose model that requires instructions, evidence and application controls.

Prompting Pattern

ROLE / TASK What should the model do? EVIDENCE What trusted information may it use? CONSTRAINTS What must it not do or claim? OUTPUT FORMAT How should the answer be structured? UNCERTAINTY What should happen when evidence is missing?
--

ENTERPRISE RULE A confident answer is not the same as a correct or authorised answer. For enterprise AI, the evidence and the authority boundary must be visible.
--

Day 2 - How RAG Works

```

PDF / Markdown documents
-> text extraction
-> chunking
-> embeddings
-> vector / semantic retrieval
-> metadata + authority filtering
-> relevant context
-> LLM
-> grounded answer + sources
  
```

Beginner Retrieval Settings

Setting	Training value	Why
Chunk size	400	Large enough for coherent context, small enough for retrieval.
Overlap	100	Preserves context across chunk boundaries.
Top K	3	Keeps the evidence packet focused and easy to inspect.
Source types	PDF or Markdown	Both represent the same synthetic corpus for training.

Embeddings - Useful Mental Model

An embedding turns text into a numeric representation so semantically related text can be compared. You do not need to visualise every dimension. The key idea is that semantic similarity helps retrieve meaningfully related content even when the exact words differ.

Keyword vs Semantic Search

Keyword search	Semantic retrieval
Best when exact terminology matters.	Best when meaning matters more than exact wording.
Easy to explain and debug.	Useful for natural-language questions.
Can miss synonyms or paraphrases.	Can retrieve conceptually similar but not authoritative content.

Day 2 - Authority, Versioning & the Nine-Document Corpus

The Procedure Trap

Version	Status	Can it be treated as current authority?
V1	SUPERSEDED	No - historical context only
V2	APPROVED / current	Yes - current procedural authority
V3D	DRAFT	No - newer does not mean approved

A Better Retrieval Decision

Semantic relevance
 + entity scope
 + approval status
 + version / effective date
 + provenance
 = trusted enterprise retrieval

CORE LESSON

Relevant does not automatically mean trusted. An enterprise RAG system should retrieve evidence, then apply authority and provenance rules before generation.

Participant Exercise

1. Ask a question about the current A-001 inspection procedure.
2. Inspect the top retrieved chunks and their source metadata.
3. Identify any DRAFT or SUPERSEDED material.
4. Apply the approval/version rule.
5. Generate an answer that cites the current approved source.
6. State what remains uncertain or requires human review.

Day 2 - RAG Evaluation & Notes

What to Evaluate

- Did retrieval return a relevant source?
- Did the application respect approval status and version?
- Does the final answer clearly distinguish source evidence from model wording?
- Are citations/provenance visible?
- Does the model avoid filling evidence gaps with invented facts?
- Does the answer preserve the human decision boundary?

Common Failure Modes

Failure	What to check
No documents found	Repository path / PDF upload path / file extensions
Wrong procedure returned	Metadata filtering and approval logic
Good source, poor answer	Prompt structure and evidence formatting
Answer has no citations	Source fields not carried into generation
Model overstates certainty	System instruction and uncertainty handling

My Day 2 Notes

DAY 3

Agentic AI & Governed Agent Patterns

Move from retrieval to controlled tool use, reusable skills, state, memory and multi-agent architecture.

Day 3 Learning Objectives

- Explain the relationship between an LLM, RAG and an agent.
- Build a bounded SQL tool and combine it with a RAG tool.
- Route a business question to the correct capability.
- Separate tools, skills, knowledge and memory.
- Use progressive disclosure to load only the instructions needed.
- Understand workflow state, tracing, permissions and human-in-the-loop controls.
- Explain when a multi-agent system is justified - and when it is not.

DAY 3 OPENING DEFINITION

An agent is an LLM-driven application that can choose approved tools to complete a bounded task. Teach one governed agent before introducing multiple agents.

Day 3 - Tools, Skills, Knowledge & Memory

Layer	Meaning	A-001 example
Tool	What the application can call	SQL query function, RAG retrieval function
Skill	How to perform a reusable capability	asset_summary.md, approved_procedure.md
Knowledge	What evidence can be retrieved	A-001 tables, nine documents
Memory	What context is intentionally retained	Current asset, last skill, unresolved evidence gap
Progressive disclosure	Load only what is needed now	Read catalog first, then selected skill only
Governance	What the system is allowed to do	Read evidence, draft briefing, stop before approval

The Three Teaching Skills

Skill	Evidence	Purpose
asset_summary	SQL only	Return current structured facts
approved_procedure	RAG only	Identify current approved guidance
reliability_review	SQL + RAG	Prepare an evidence-backed human-review briefing

Progressive Disclosure

Question

- > read small skills catalog
- > choose one skill
- > load only that skill's instructions
- > expose only the allowed tools
- > gather evidence
- > produce bounded output

Day 3 - Agent Routing & Evidence

A Question RAG Alone Cannot Answer

EXAMPLE QUESTION

What was happening in A-001's data, what does the current approved procedure say, and what should be prepared for human reliability review?

User question

- | | |
|----------------------|--|
| -> Agent / Router | |
| -> SQL tool | -> current structured facts |
| -> RAG tool | -> approved document evidence |
| -> Analysis | -> bounded calculations / interpretation |
| -> Evidence packet | |
| -> Bounded synthesis | |
| -> Human review | |

Tool Contract Checklist

- Allowed input
- Allowed data / document scope
- Expected output
- Failure behaviour
- Authority boundary
- Provenance / audit fields

DESIGN PRINCIPLE

The orchestrator may choose a tool. It may not expand the tool's permissions. A smarter planner does not justify weaker controls.

Day 3 - Memory, State, Governance & Multi-Agent Systems

Memory vs Workflow State

Memory	Workflow state
Retained context that may help future steps or turns.	The explicit stage of the current process.
Example: selected asset, last skill, unresolved evidence gap.	Example: RECEIVED -> GUARDED -> INVESTIGATED -> DRAFTED -> REVIEWED.
Should be intentional and bounded.	Should be inspectable, attributable and auditable.

When Multi-Agent Makes Sense

Use multiple agents when...	Prefer one agent + tools when...
Different roles require different permissions or specialist capabilities.	The workflow is small and the same authority boundary applies.
Independent traces or ownership boundaries matter.	Extra agents would add complexity without improving control.
A supervisor must coordinate distinct specialist outputs.	One router can choose among a few bounded tools.

Reference Pattern

```
Supervisor
-> Data specialist
-> Knowledge specialist
-> Analysis specialist
-> Reporting specialist
    -> Human review
```

INSTRUCTOR REFERENCE

Multi-agent is an architecture choice, not a maturity badge. If one governed agent plus tools is enough, that is usually the better teaching and engineering choice.

Day 3 - Hands-on Checklist & Notes

Done	Task
☐	Run the bounded SQL tool for A-001
☐	Run the governed RAG tool
☐	Ask a hybrid question requiring SQL + RAG
☐	Inspect which tool/skill was selected
☐	Load only the selected skill instructions
☐	Inspect memory / workflow state
☐	Identify the human-approval boundary
☐	Sketch whether a multi-agent design is justified

My Day 3 Notes

DAY 4 - FIRST HALF

Integrated Enterprise AI Application

Combine structured data, ML, RAG, agent tools and governance into one inspectable workflow.

Integration Architecture

USER

-> Enterprise AI Application

-> Governed Orchestrator

-> Structured Data

-> RAG / Documents

-> ML Output

-> Evidence Packet

-> Draft Briefing

-> Human Review

-> Recorded Outcome

What Must Stay Separate

Evidence type	Example	Why separation matters
Observed structured facts	health score, work-order count	Directly queryable facts
Sensor / model evidence	trend, forecast, MAE	Analytical evidence - not a diagnosis
Written guidance	approved V2 procedure	Organisational authority
Model interpretation	summary or recommendation	Generated reasoning; must be traceable
Human decision	approve / revise / reject	Final authority

DAY 4 RULE

Introduce very little new theory. The first half is about integration, traceability and rehearsal for the capstone.

Team Formation Sheet

Form teams before the capstone if possible. Roles are flexible and can be combined in smaller teams. A useful team size is whatever allows everyone to contribute without creating coordination overhead.

Suggested Roles

Role	Primary responsibility
Team facilitator	Keep the team on the business question and timebox decisions.
Data lead	Structured-data exploration and ML evidence.
Knowledge / RAG lead	Document authority, retrieval and citations.
Agent / integration lead	Tool routing, evidence packet and workflow controls.
Reviewer / presenter	Challenge assumptions, check limitations and lead the final demo.

Team Assignments

Team	Members	Primary roles	Working title	Instructor notes
Team 1				
Team 2				
Team 3				
Team 4				
Team 5				
Team 6				
Team 7				
Team 8				

INSTRUCTOR REFERENCE

Create teams before the final half-day where possible. During the capstone, avoid solving the project for them; ask evidence and boundary questions instead.

DAY 4 - SECOND HALF

Team Capstone: A-001 Condition / Reliability Review

Use the full stack to prepare an evidence-backed decision-support briefing and demonstrate the workflow.

Capstone Scenario

An engineer has observed unusual behaviour around synthetic Asset A-001. Your team must prepare an evidence-backed condition/reliability review using the available enterprise data and knowledge.

Business Question

CAPSTONE QUESTION

What does the available evidence say about A-001, what should a qualified reviewer pay attention to, and what is the appropriate human next step?

Required Evidence

- structured enterprise data
- recent sensor trend and/or ML output
- approved document guidance
- work-order / inspection / field context where relevant
- clear provenance and source identifiers
- evidence gaps, conflicts and uncertainty

Boundary

THE CAPSTONE OUTPUT IS DECISION SUPPORT

The application may prepare evidence and a review briefing. It must not diagnose a real physical failure, issue control commands, authorise maintenance, close work orders, treat a model output as approval, or approve its own recommendation.

Capstone Workflow

1. Understand and restate the business question.
2. Inspect the relevant structured data.
3. Review the recent sensor trend and ML/forecast output.
4. Retrieve relevant enterprise documents.
5. Confirm which procedure/policy is currently authoritative.
6. Combine evidence without blurring source types.
7. Identify conflicts, limitations and missing evidence.
8. Prepare a bounded review briefing.
9. Stop at the human-review boundary.
10. Prepare a concise team demonstration.

Suggested Half-Day Flow

Stage	Suggested time	Outcome
Kickoff & role assignment	20 min	Business question, team roles, success criteria
Evidence gathering & build	100 min	SQL, ML, RAG and agent evidence
Validation & challenge	30 min	Check sources, authority, gaps and governance
Demo preparation	30 min	Clear story, evidence packet, limitations
Team demonstrations	Remaining time	Short demo + questions / feedback

ADAPT TO CLASS SIZE

The timing is a suggested structure, not a rigid schedule. Protect enough time for teams to explain evidence, limitations and human oversight - not just show a UI.

Capstone Output Template

1. Situation

What triggered the review?

2. Structured evidence

What do the database, work and sensor records show?

3. Predictive / analytical evidence

What does the ML/forecast output suggest, and what does it NOT prove?

4. Document evidence

Which approved documents are relevant? Include IDs, versions and status.

5. Assessment

What can reasonably be concluded from the combined evidence?

6. Evidence gaps / uncertainty

What is unknown, conflicting or unverified?

7. Recommended human next step

What should a qualified reviewer inspect, confirm or decide next?

8. Sources used

List the tables/views and document IDs used.

Capstone Evaluation Rubric

Dimension	1-5: What good work demonstrates
Problem understanding	Frames the business question correctly and avoids jumping straight to a technical answer.
Data use	Uses structured data appropriately and separates observations from interpretation.
AI / RAG use	Retrieves relevant knowledge and respects document approval/version status.
Evidence & explainability	Shows where important claims came from and identifies limitations.
Governance & human oversight	Preserves uncertainty and keeps final operational authority with a human reviewer.

Score Sheet

Dimension	Score / 5	Comments
Problem understanding		
Data use		
AI / RAG use		
Evidence & explainability		
Governance & human oversight		
TOTAL	/ 25	

TRAINING INTERPRETATION

21-25: strong end-to-end understanding | 16-20: good working understanding | 11-15: partial understanding | 5-10: requires guided remediation. This is training feedback, not an operational competency certification.

Final Demonstration Checklist

Done	Your team should be able to show...
☐	The business question in one sentence
☐	The structured data / sensor evidence used
☐	The ML or analytical output and its limitation
☐	The approved document evidence and source IDs
☐	The tool / routing path used by the application
☐	The final bounded briefing
☐	Uncertainty or evidence gaps
☐	Where the human reviewer remains responsible

Suggested Demo Story

1. Problem
2. Evidence
3. What the AI application did
4. What it found
5. What it cannot conclude
6. What a qualified human should review next

Questions the Instructor May Ask

- Why did you trust this document and not the other version?
- Which part of your answer came from data and which part came from written guidance?
- What would happen if retrieval returned no eligible approved evidence?
- Why is your agent allowed to call this tool?
- What decision is deliberately left to a person?
- What would you improve before this architecture could be considered production-ready?

Quick Reference: Core Concepts

Concept	One-line definition
Delta table	A governed lakehouse table used for reliable analytics and downstream AI.
Feature	An input variable used by a model.
Target	The value a supervised model learns to predict.
MAE	Average absolute prediction error in the target's units.
LLM	A language model that generates text from instructions and context.
RAG	Retrieval plus generation: supply enterprise evidence before asking the model to answer.
Chunk	A smaller unit of document text used for retrieval.
Embedding	Numeric representation used to compare semantic meaning.
Metadata	Context such as document ID, version, status, date or entity scope.
Tool	A callable capability such as SQL or document retrieval.
Skill	Reusable instructions describing how a capability should be performed.
Memory	Intentionally retained context across steps or turns.
Progressive disclosure	Load only the detailed instructions needed for the selected task.
Agent	An LLM-driven application that selects approved tools to complete a bounded task.
Multi-agent system	Multiple specialised agents coordinated when roles, permissions or audit boundaries justify it.
Human-in-the-loop	A deliberate point where a person reviews, approves, revises or rejects an output.
Provenance	Evidence showing where a claim or output came from.

Troubleshooting & Resource Paths

Symptom	What to check
Database not found	Exact V2.2 filename and Volume path
Tables not visible	Setup completed; refresh Catalog; confirm schema
a001_sensor_hourly missing	Confirm V2.2 database was imported
RAG finds no documents	Repository document path or original PDF upload path
Wrong procedure treated as current	Approval-status / version filtering
Skills not found	Repo skills path or fallback Workspace folder
Model endpoint error	Workspace entitlement / endpoint availability
Agent overreaches	Tool scope, system constraints and human-review boundary
Demo takes too long	Return to pre-staged environment and protect the learning objective

Important Paths

Database:
 /Volumes/workspace/nuclear_enterprise_360/training_files/
 nuclear_enterprise_360_v2_2_clean.db

Repo-native A-001 documents:
 Day_2_GenAI_RAG/documents/a001/

Original PDF fallback:
 /Workspace/Nuclear_Enterprise_360/A001 Documents/

Repo-native agent skills:
 Day_3_Agentic_AI/skills/

Skill fallback:
 /Workspace/Nuclear_Enterprise_360/02_Agent_Skills/

Useful Repository Areas

Folder	Purpose
Day_1_Databricks_Data_ML	Setup, SQL, forecasting and optional anomaly notebooks
Day_2_GenAI_RAG	RAG notebooks, document corpus and evaluation files
Day_3_Agentic_AI	Agent notebooks and skill files
Day_4_Integrated_Capstone	Integration notebook, capstone brief and rubric
data	Data dictionary, quick-start SQL and validation
resources	Participant setup and programme references

Protect these five outcomes in order: (1) Day 1 setup works, (2) one forecasting notebook works, (3) one RAG notebook works, (4) one SQL + RAG agent works, (5) one integrated capstone path works.

From	Transition question	Leads to
Data exploration	Can structured facts alone explain what the organisation expects?	ML / knowledge context
Forecasting	Is prediction enough to make a maintenance decision?	Day 2 RAG
RAG	What if a question needs both current database facts and approved written guidance?	Day 3 agents
Single agent	When would specialised roles and permissions justify multiple agents?	Multi-agent architecture
Integration	Can the system show evidence, limitations and the human boundary end-to-end?	Capstone

We started with a table of sensor readings. We ended with an AI-enabled application that can understand structured data, make a prediction, retrieve trusted organisational knowledge, select bounded tools, assemble evidence, preserve uncertainty and keep a human reviewer in control.

Final Participant Checklist

I can...	Check
Explain the Databricks data hierarchy used in the course.	☐
Query A-001 using SQL.	☐
Explain the difference between a feature, target, forecast and MAE.	☐
Explain why a forecast is evidence, not a maintenance decision.	☐
Explain prompts, tokens, context and hallucination risk.	☐
Explain chunking, embeddings and semantic retrieval.	☐
Identify the current approved A-001 procedure.	☐
Explain why newest / most similar is not always authoritative.	☐
Explain tools vs skills vs knowledge vs memory.	☐
Describe progressive disclosure.	☐
Explain when one agent is enough and when multi-agent may be justified.	☐
Describe the human-approval boundary in the A-001 workflow.	☐
Demonstrate the integrated capstone story with evidence and limitations.	☐

My Most Important Takeaways

[illegible]

Notes 1

Notes 2